Grocery Delivery System

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Problem Statement

While buying groceries may seem simple, for many people it has become a frustrating task. People are busy, stores are crowded, and many do not have the time to shop in person. Older customers are one example of the kinds of people who struggle to complete their own shopping, whether it's because of heavy items or long wait times in store. My project will focus on solving these problems for customers by creating a grocery delivery system that is simple to use and reliable.

System Requirements

This system will allow users to do the following:

- Create an account.
- Log into it securely.
- Search for grocery items by name or category.
- Add items to a shopping cart.
- Update quantities.
- Place orders.
- Select delivery time slots.
- View past orders.
- Track current deliveries.
- Receive notifications about order confirmation and delivery status.
- Store managers can add, update, or remove products.
- Store managers can manage inventory and customer orders.

Functional Requirements

No.	Priority Weight	Description
REQ-1	High	The system shall allow customers to create a user account

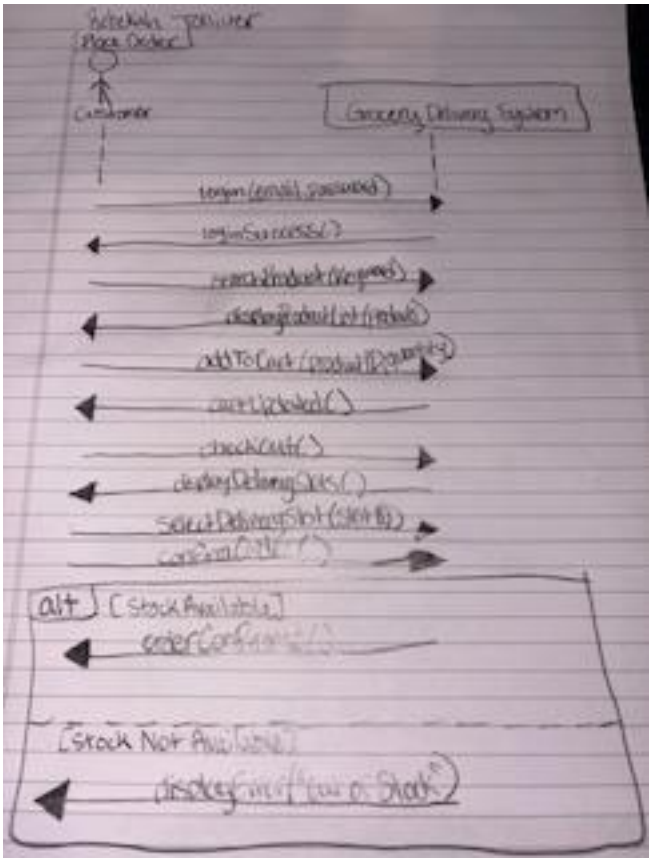
REQ-2	High	The system shall allow customers to log in securely
REQ-3	High	Customers shall be able to search grocery items by name or category
REQ-4	High	Customers shall be able to add products to a shopping cart
REQ-5	High	Customers shall be able to update or remove items from the shopping cart
REQ-6	High	Customers shall be able to place an order
REQ-7	Medium	Customers shall be able to select a delivery time slot
REQ-8	Medium	Customers shall be able to view their past orders
REQ-9	High	Customers shall be able to track the status of current deliveries
REQ-10	Medium	Customers shall receive notifications for order confirmation and delivery updates
REQ-11	High	Store managers shall be able to add, update, or remove products
REQ-12	High	Store managers shall be able to manage inventory levels
REQ-13	Medium	Store managers shall be able to view and manage customer orders
REQ-14	Medium	Delivery staff shall be able to update delivery status

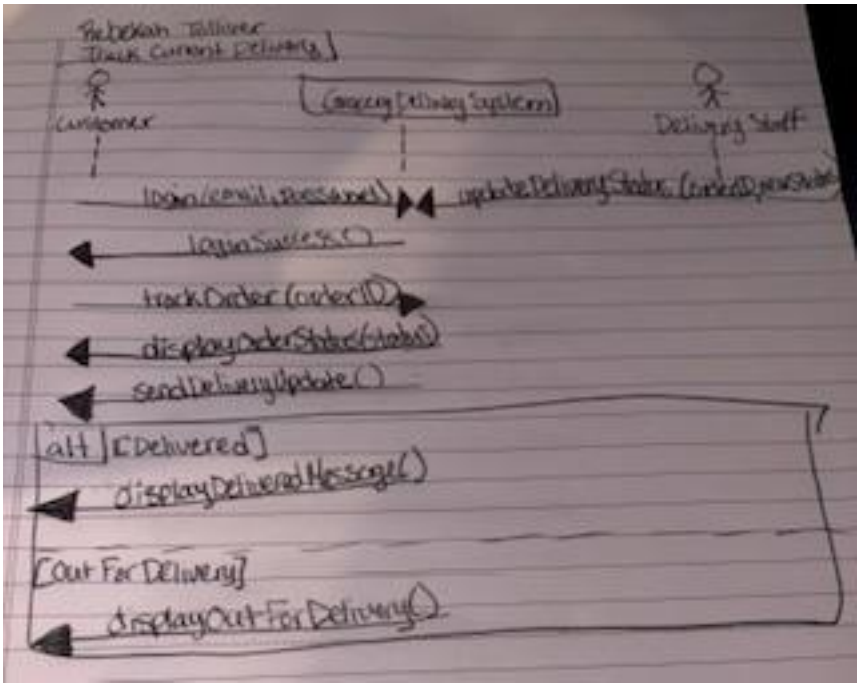
Nonfunctional Requirements:

No.	Category	Priority	Description
NFR-1	Usability	High	The system shall be easy to use for elderly and non-technical users
NFR-2	Reliability	High	The system shall correctly save orders and inventory data without loss
NFR-3	Performance	Medium	Pages shall load within a reasonable time under normal usage
NFR-4	Functionality	Medium	The system shall provide accurate order tracking and notifications

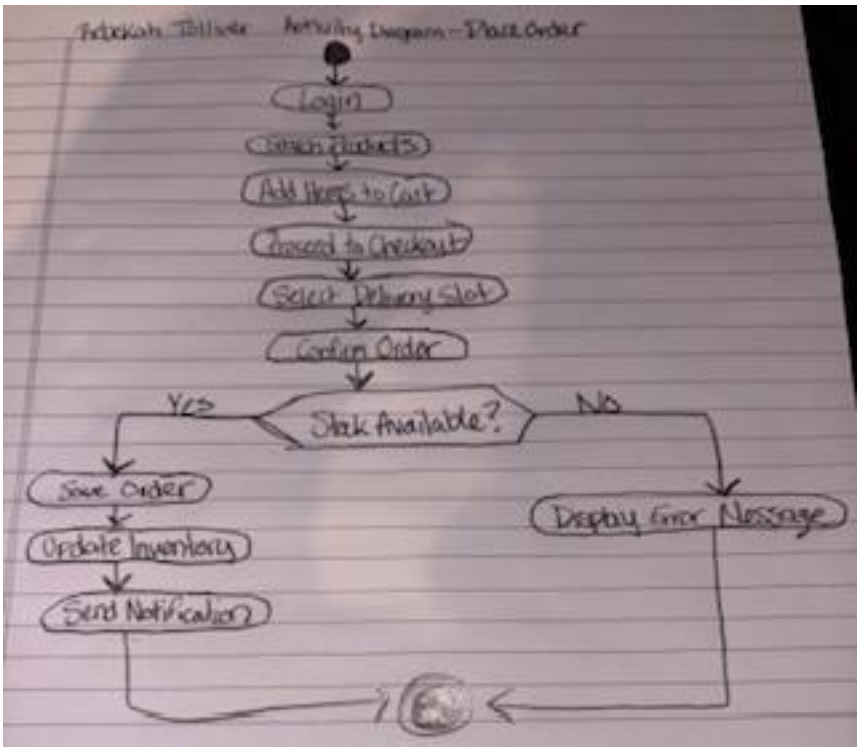
NFR-5	Supportability	Low	The system shall allow future updates and feature additions

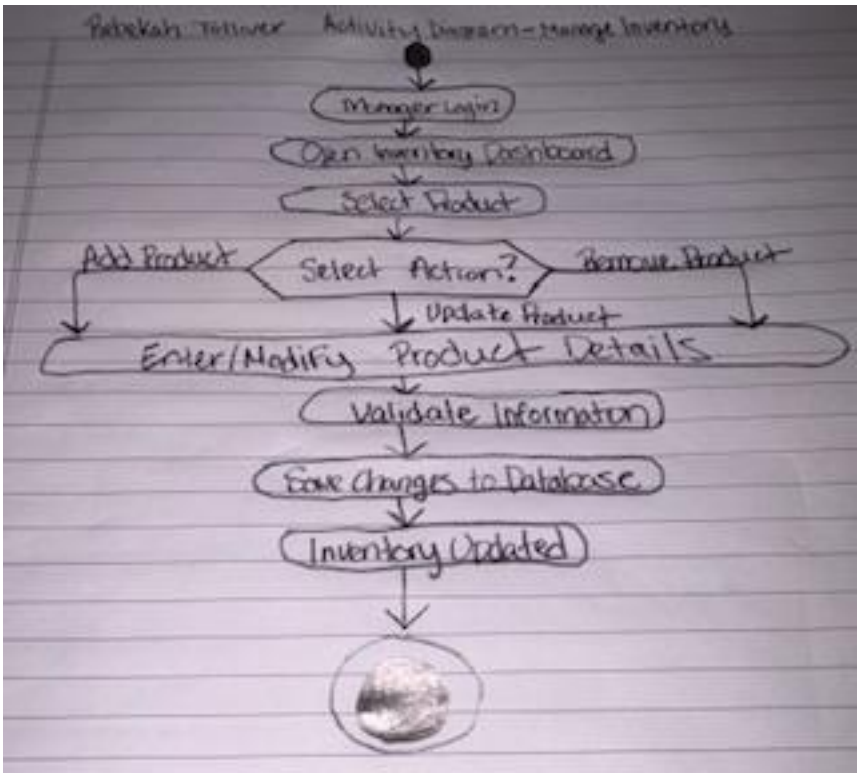
System Sequence Diagrams





Activity Diagrams





User Interface Specifications

Selected Use Cases:

- Create Account (Customer)
- Place Order (Customer)
- Track Current Delivery (Customer + Delivery Staff)
- Add / Update / Remove Products (Store Manager)
- Manage Inventory (Store Manager)

1. Create Account (Customer)

How Users Enter Information:

- User clicks “Register” on the homepage.
- Enters name, email, password, phone number, and address in clearly labeled text fields.
- Clicks the “Create Account” button.

How Results are Displayed:

- Success: “Account created successfully. Please log in.”
- Error: “Email already registered” or “Password too short.”

Navigation Path:
Homepage > Register > Fill Form > Create Account > Confirmation Message

User Effort Estimation:

- Clicks: Register (1) + name field (1) + email field (1) + password field (1) + phone field (1) + address field (1) + Create Account button (1) = 7 clicks
- Keystrokes: Name (10) + Email (15) + Password (8) + Phone (10) + Address (20) = 63 keystrokes

2. Place Order (Customer)

How Users Enter Information:

- Login to account.
- Search for products via search bar or category selection.
- Click “Add to Cart” for each product.
- Modify quantities or remove items in the cart.
- Click “Proceed to Checkout”.
- Select delivery time slot.
- Click “Confirm Order”.

How Results are Displayed:

- Confirmation message: “Order placed successfully. Your delivery is scheduled for [time slot].”
- Error message: “Item out of stock.”

Navigation Path:
Login > Product Search / Browse > Add to Cart > Shopping Cart > Checkout > Select Delivery Slot > Confirm Order > Confirmation

User Effort Estimation:

- Clicks: Login (1) + Search Bar (1) + Select Product (1) + Add to Cart (1) + Cart (1) + Modify / Remove (1) + Proceed to Checkout (1) + Delivery Slot (1) + Confirm Order (1) = 9 clicks
- Keystrokes: Search term (10), Quantity (2) = 12 keystrokes

3. Track Current Delivery (Customer + Delivery Staff)

How Users Enter Information:

- Customer logs in.
- Clicks “Track Order”.
- System retrieves current status and displays: “Out for Delivery”, “Delivered”, or estimated arrival time.
- Delivery Staff logs in.
- Updates delivery status for assigned orders via a drop-down or button selection.

How Results are Displayed:

- Customer sees real-time updates: map view (optional) or status text.
- Delivery Staff receives confirmation of status update.

Navigation Path:
Customer: Login > Track Order > View Status
Delivery Staff: Login > Assigned Orders > Update Status > Confirmation

User Effort Estimation (Customer):

- Clicks: Login (1) + Track Order (1) + View Status (1) = 3 clicks
- Keystrokes: None required

User Effort Estimation (Delivery Staff):

- Clicks: Login (1) + Assigned Orders (1) + Update Status (1) + Confirm (1) = 4 clicks
- Keystrokes: None required

4. Add / Update / Remove Products (Store Manager)

Add Product:

- Navigate to Manager Dashboard.

- Click “Add Product”.
- Enter product name, category, price, stock quantity, description.
- Click “Save Product”.

Update Product:

- Click “Edit” next to product.
- Modify fields.
- Click “Save Changes”.

Remove Product:

- Click “Delete” next to product.
- Confirm deletion in pop-up modal.

How Results are Displayed:

- Success message: “Product added/updated/removed successfully.”
- Error messages for invalid data or conflicts.

Navigation Path:
Login > Manager Dashboard > Products > Add / Edit / Delete > Confirmation

User Effort Estimation (Add Product):

- Clicks: Dashboard (1) + Add Product (1) + Name field (1) + Category field (1) + Price field (1) + Quantity field (1) + Description field (1) + Save Product (1) = 8 clicks
- Keystrokes: Name (10) + Category (8) + Price (5) + Quantity (2) + Description (15) = 40 keystrokes

5. Manage Inventory (Store Manager)

How Users Enter Information:

- Manager logs in.
- Opens inventory dashboard.
- Selects product.
- Updates stock quantity or status.

- Clicks “Save Changes”.

How Results are Displayed:

- Success: “Inventory updated successfully.”
- Error: “Quantity must be numeric.”

Navigation Path:
Login > Manager Dashboard > Inventory > Select Product > Update > Save Changes > Confirmation

User Effort Estimation:

- Clicks: Login (1) + Dashboard (1) + Inventory (1) + Select Product (1) + Quantity Field (1) + Save Changes (1) = 6 clicks
- Keystrokes: Quantity (2) = 2 keystrokes

Project Plan

Software: HTML, CSS, JavaScript

Hardware: Any PC / laptop

Network: Standard internet connection

Development Timeline:

- Week 1–2: Requirements finalized, project setup
- Week 3–4: Login, registration, basic browsing
- Week 5–7: Cart and checkout
- Week 8: Mid-term demo
- Week 9–11: Order tracking, notifications
- Week 12–14: Bug fixes, performance improvements

- Week 15: Final testing, presentation

References

Course Material

HTML, CSS, and JavaScript from my Web Design and Development course